

Design and Analysis of Control Strategies for Vehicle Platooning

Apoorva Saxena¹, Hong Li², Dip Goswami¹, Chetan Belagal Math¹

¹Eindhoven University of Technology, Eindhoven, Netherlands

²Car Infotainment & Driving Assistance, NXP Semiconductors, Eindhoven, Netherlands

Email: a.saxena@student.tue.nl, hong.r.li@nxp.com, d.goswami@tue.nl, c.belagal.math@tue.nl

Abstract—This paper presents a novel vehicle platoon control algorithm using Vehicle to Vehicle (V2V) and Vehicle to Infrastructure (V2I) wireless communications between platoon members. A platoon forms a chain of vehicles (e.g., trucks) for improved traffic and fuel efficiency. Platooning algorithms aim to maintain small inter-vehicular distance between vehicles under dynamic driving conditions like acceleration and deceleration. The random delay and dropped messages have a pronounced impact on the control algorithm especially in a congested communication network. In this work, a decentralized platooning control strategy is proposed such that each vehicle can independently switch between multiple controller strategies dependent on communication topologies or the pattern of information exchange from vehicles in-front. The control strategy takes into account V2V communication delay of up to 100ms. In order to overcome the impact of dropped messages a predictive control scheme is proposed. We show the existence of well-known Common Quadratic Lyapunov Function (CQLF), to prove the stability of this switched system. Simulation results show that the proposed control algorithm maintains stability and can keep the inter-vehicle gap deviation less than $\pm 6.3\%$ under imperfect communication conditions for a sinusoidal acceleration input between $\pm 2\text{m/s}^2$.

Index Terms—CACC, Predecessor Follower, Pre-Predecessor follower, Leader follower, switching topology, vehicle platooning, V2V

I. INTRODUCTION

Nowadays traffic safety and efficiency have become very serious problems. Global status report on Road safety- 2015 by WHO indicates total death toll of almost 1.25 million per year in road accidents, while the number of light duty vehicles will grow to 2 billion in 2050 compared to 850 million in 2013 [2]. Intelligent Transportation Systems (ITS) applications, including vehicle platooning technology are promising to improve on-road safety, effective road usage and reducing fuel consumption.

By detecting or sharing predecessor information using radar sensors or V2V and V2I wireless communications, vehicles can travel together in an actively coordinated fashion like road trains, known as platoons. The main objective of vehicles in a string of a platoon is to ensure constant and safe inter-vehicle distance along with successful tracking of velocity and acceleration of its predecessor. A platooning algorithm utilizes a control strategy to compute the right acceleration of the member vehicles to achieve these objectives and the performance of a platooning controller further depends on the communication network topology.

Adaptive Cruise Control (ACC) technologies uses radar or laser sensors to follow its direct predecessor [9]. One problem of ACC is the one-by-one propagation of shockwaves

under dynamic conditions like an emergency braking. Therefore, it requires to maintain a large inter-vehicle distance for safety and comfort [6]. A large inter-vehicle distance cannot sufficiently reduce air flow resistance to improve fuel efficiency.

Cooperative adaptive cruise control (CACC) extends the capability of ACC by exchanging information over V2V and V2I communication, e.g. via Dedicated Short Range Communication (DSRC) protocols. In CACC a vehicle receives information not only from its predecessor but also from other members of the platoon. Richer information communication between vehicles opens up the possibility of improving the performance of ACC. A CACC system provides much stable and cooperative control operations by limiting propagation of disturbances throughout the vehicle chain, and thus it has potential to reduce inter-vehicle distance.

However, wireless communication imperfection plays a vital role in CACC and it becomes crucial particularly under network congestion. Relevant questions are: What if information from other platoon members gets delayed or lost due to heavy network load or bad channel condition? Whether system stability can be maintained under congested network conditions? Various recent studies have provided promising results considering network induced delay. However most research attempts to mitigate its impact through a given communication network topology [1], [3]. Most common communication topology is Predecessor Follower where each platoon member follow its predecessor. Leader-Predecessor Follower, Leader Follower etc. are few others network topologies. Each topology has its own advantages and disadvantages. The overall platooning performance usually suffers depending on the choice of the network topology. Often, a larger inter-vehicle distance is required for achieving stable operation of a platoon for a given network topology.

In this paper, we propose a novel control strategy through intra-platoon information management that utilizes information from multiple vehicles ahead by means of switching between multiple communication topologies. Switching occurs to mitigate the impact of loss of information from one vehicle, by utilizing the information from other vehicles ahead to obtain a stable and safer operation. Through this approach we show that string stability can be achieved for shorter inter-vehicular distance under imperfect network conditions.

The rest of the paper is structured as follows. Section II presents work related to platooning and the state of the art on CACC. Section III presents basic longitudinal vehicle model

with respect to multiple network topologies. Section IV presents the closed loop platoon system model with network delay along with the stability analysis and simulation results of individual topologies. Section V describes proposed control strategy and its stability analysis through CQLF. Section VI presents simulation results. Finally, Section VII makes concluding remarks and future work.

II. RELATED WORK

In the context of vehicle platooning and control, ACC and CACC are two main technology ingredients. In ACC, a vehicle automatically follows the predecessor velocity to maintain a desired distance by means of sensing predecessors states through radar or laser. However under dynamic conditions i.e. due to any variations in the velocity of leader vehicle disturbances are propagated throughout the string [9]. For safety reasons, the ACC requires a larger inter-vehicle distances [6]. For example, as per International Organization for Standardization (2010) a minimum of 1 sec time gap (time required by a vehicle to reach to its predecessor position at its current velocity) is required in ACC. For a higher traffic efficiency and throughput, it is necessary to reduce the time gap [7]. A shorter time gap can potentially improve fuel efficiency of vehicle by reducing aerodynamic drag, in particular heavy duty vehicles [8]. These features are the highlights of Cooperative Adaptive Cruise Control (CACC).

Various studies like [1], [3] have proposed the controller that uses traditional ACC for vehicle following objective and utilize predecessor's acceleration information exchanged over DSRC in the feed-forward fashion to improve tracking performance. Effects of communication delay in CACC have been widely studied in literature [10]-[14]. These works use either predecessor follower topology (PF) or Leader-Predecessor (LP) follower topology [12] [14] to mitigate the effects of communication delay.

Predecessor follower is the most commonly used network topology as the probability to receive information over the network from predecessor is the highest. However in PF any disturbance in the platoon leader gets amplified throughout the chain which gets worse under communication delay. Also each vehicle is only dependent on its predecessor information. What if under congested network this information is lost for consecutive periods? Works using LP topology make use of weighted average approach. By utilizing information from leader and predecessor each vehicle instantly reacts to any disturbance in the leader. The vehicle doesn't have to wait for its predecessor's reaction. However studies done in [12] have considered these weights as design parameters that changes with position of a platoon member with respect to its leader. There is no algorithm proposed to dynamically obtain these weights. Also for a vehicle far away from leader has the least probability to receive its information. What if information from leader and predecessor are lost simultaneously?

In this paper we have proposed a control algorithm that utilizes the information from multiple vehicles in-front by means of switching between multiple topologies. Along with

this a predictive control strategy is proposed to maintain the stability under consecutive periods of message loss from vehicles in-front due to time-varying communication link failure. Each vehicle in the platoon switches independently on the basis of information it has received through wireless communication from vehicle in-front. By means of showing the existence of CQLF, stability of the algorithm is analyzed under the switching communication network topologies.

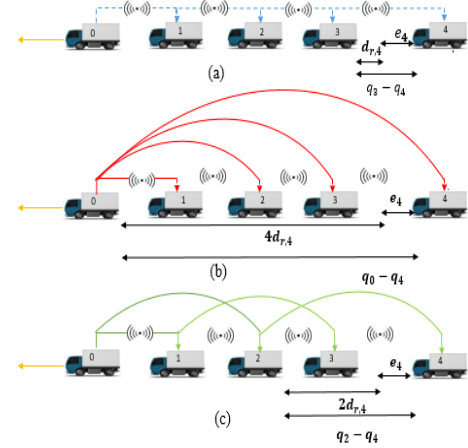


Fig. 1: Multiple Communication Network Topologies: a) Predecessor follower; b) Leader follower; c) Pre-predecessor follower

III. BASIC PLATOON MODELS AND SETTINGS

In this paper we have articulated three core communication topologies i.e. Predecessor follower (PF), Pre-Predecessor Follower (PPF) and Leader Follower (LF). As shown in Fig. 1, desired distance of the i^{th} vehicle varies with respect to vehicle it is following.

Here we consider constant time headway spacing strategy where desired inter vehicular distance ($d_{r,i}$) between a vehicle and its predecessor is given by

$$d_{r,i} = r_i + h_{d,i} \Delta v_i \quad (1)$$

where i is the vehicle index, r_i is the constant buffer gap between vehicles, $h_{d,i}$ is the constant headway time-gap constant i.e., time taken by i^{th} vehicle to reach its predecessors position when $r_i = 0$ and $\Delta v_i = v_i - v_k$ is the vehicles relative velocity with respect to predecessor or pre-predecessor or leader depending on communication topology. All the assumptions taken with respect to modelling are shown in Table I.

The inter-vehicular distance for different topologies is given by,

$$d_{PF,i} = q_{i-1} - q_i \quad (2)$$

$$d_{PPF,i} = q_{i-2} - q_i^\dagger \quad (3)$$

$$d_{LF,i} = q_0 - q_i \quad (4)$$

where q_i is the absolute i^{th} vehicles position. So the control objective for ensuring constant inter-vehicle distance, as shown in Fig. 1 for multiple topologies is to nullify the position error given by,

$$e_{PF,i} = q_{i-1} - q_i - (r_i + h_{d,i} \Delta v_i) \quad (5)$$

$$e_{PPF,i} = q_{i-2} - q_i - 2(r_i + h_{d,i}\Delta v_i)^\dagger \quad (6)$$

$$e_{LF,i} = q_0 - q_i - (i)(r_i + h_{d,i}\Delta v_i) \quad (7)$$

where $\Delta v_i = v_i - v_k$, here index $k = i-1, i-2, 0$ for PF, PPF and LF topology respectively.

A. Basic Vehicle Longitudinal dynamics

The linearized longitudinal dynamics of each vehicle in the string is given by [3]:

$$\dot{q}_i(t) = v_i(t) \quad (8)$$

$$\dot{v}_i(t) = a_i(t) \quad (9)$$

$$\dot{a}_i(t) = -\eta_i^{-1}a_i(t) + \eta_i^{-1}u_{i,des}(t) \quad (10)$$

Where η_i internal dynamics actuator time constant and $u_{i,des}$ is the desired acceleration or control input of the i^{th} vehicle. In above equations $q_i(t), v_i(t), a_i(t)$ are absolute position, velocity and acceleration respectively.

B. Control Structure

In this paper we consider a hierarchical control structure [15] consisting of two levels as shown in Fig. 2.

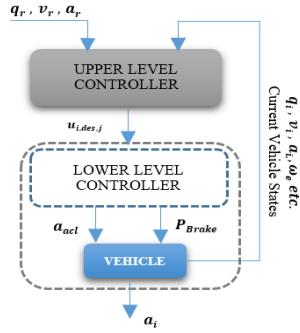


Fig. 2: Hierarchical control architecture

The upper level controller acts on inter-vehicular states received via network and determines the desired acceleration ($u_{i,des,j}$). In this layer, a PD-controller is used to perform vehicle following objective (in complex frequency domain):

$$u_{i,des,j}(s) = \omega_{k,i}(\omega_{k,i} + s)E_j(s) \quad (11)$$

where index j is for either PF, LF or PPF topology, $\omega_{k,i}$ is the bandwidth of controller such that $\omega_{k,i} \ll \omega_{G,i} = 1/\eta_i$ [3] to prevent actuator saturation and $E_j(s)$ is the position error in frequency domain. Thus by substituting values of error from (5), (6) and (7) in time domain equivalent of (11) for each topology desired acceleration is given by:

$$u_{i,des,j}(t) = k_{p,i}e_{j,i} + k_{d,i}\dot{e}_{j,i} \quad (12)$$

where $k_{p,i} = \omega_{k,i}^2$ and $k_{d,i} = \omega_{k,i}$.

Next, a lower level controller takes into account all the nonlinearities coming from engine, aerodynamic drag and friction etc. and it further determines pedal position (a_{acl}) and actual break position (p_{break}). Combination of the lower level controller and the vehicle forms a plant that is modelled

[†]Note- In PPF vehicle at $i=1$ follows equation (2) and (5) respectively as only vehicle at $i=0$ is in-front

by Generalized Vehicle Longitudinal Dynamics (GVLD) [15] with input as desired acceleration and output as actual acceleration,

$$a_i = \frac{K_f}{\eta_i s + 1} u_{i,des,j} \quad (13)$$

where K_f is the system gain and η_i is internal dynamics actuator time constant. In this work, we mainly focus on the upper level control assuming a perfect lower level controller in place.

C. CACC vehicle model

Equations (8), (9) and (10) can be lumped and written in the form of state space representation as:

$$\dot{x}_i(t) = A_i x_i(t) + B_{s,i} u_{i,des,j}(t) \quad (14)$$

where $[x_i]^T = [q_i \ v_i \ a_i]$ is the state vector, A_i is system matrix, $B_{s,i}$ is input matrix given by:

$$A_i = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -\eta_i^{-1} \end{bmatrix}, B_{s,i} = \begin{bmatrix} 0 \\ 0 \\ \eta_i^{-1} \end{bmatrix}$$

In equation (14) $u_{i,des,j}$ is the desired acceleration or control input that can be derived separately for different topologies by substituting error from (5), (6) and (7) in (12) and can be rewritten in state space representation as given below:

For PF topology:

$$u_{i,des,PF} = k_{i,i-1}x_{i-1} + k_{i,i}x_i + k_{c,i} \quad (15)$$

$$k_{i,i-1} = \begin{bmatrix} k_{p,i} \\ k_{p,i}h_{d,i} + k_{d,i} \\ k_{d,i}h_{d,i} \end{bmatrix}^T, k_{i,i} = \begin{bmatrix} -k_{p,i} \\ -k_{p,i}h_{d,i} - k_{d,i} \\ -k_{d,i}h_{d,i} \end{bmatrix}^T$$

and

$$k_{c,i} = [-k_{p,i}r_i]^T$$

For PPF topology:

$$u_{i,des,PPF} = k_{i,i-2}x_{i-2} + k_{i,i}x_i + k_{c,i} \quad (16)$$

$$k_{i,i-1} = \begin{bmatrix} k_{p,i} \\ k_{p,i}h_{d,i} + k_{d,i} \\ k_{d,i}h_{d,i} \end{bmatrix}^T, k_{i,i} = \begin{bmatrix} -k_{p,i} \\ -2k_{p,i}h_{d,i} - k_{d,i} \\ -2k_{d,i}h_{d,i} \end{bmatrix}^T$$

and

$$k_{c,i} = [-2k_{p,i}r_i]^T$$

For LF topology:

$$u_{i,des,LF} = k_{i,0}x_0 + k_{i,i}x_i + k_{c,i} \quad (17)$$

$$k_{i,i-1} = \begin{bmatrix} k_{p,i} \\ k_{p,i}h_{d,i} + k_{d,i} \\ k_{d,i}h_{d,i} \end{bmatrix}^T, k_{i,i} = \begin{bmatrix} -k_{p,i} \\ -ik_{p,i}h_{d,i} - k_{d,i} \\ -ik_{d,i}h_{d,i} \end{bmatrix}^T$$

and

$$k_{c,i} = [-ik_{p,i}r_i]^T$$

Parameters	Assumptions
Number of vehicles (n)	5
Constant Buffer Distance (r)	3 m
Headway time gap constant (h_d)	0.1 sec
Internal Dynamics Actuator Time Constant (η_i)	0.25 sec
Acceleration	$-2 \text{ m/s}^2 < a_f < 2 \text{ m/s}^2$
K_p, K_d	0.5, 1.8
Motion Control	Longitudinal Control
Sampling Time (h)	100 ms
Time-delay	100 ms

TABLE I: Parameters for platooning

Depending on the topology, the i^{th} vehicle in a platoon receives the states x_{i-1} , x_{i-2} and x_o from the predecessor, pre-predecessor and the leader respectively, in form of packets through wireless communication network. The imperfections introduced by the network induces a certain amount of *time-delay* in the reception of these packets that impacts the performance of the controller. To overcome its impact, we have taken into account a *time-delay* of upto 100ms such that all vehicles simultaneously update their states using input delayed upto 100ms from predecessor, pre-predecessor or leader. A delay of 100ms is based on the 10Hz Common Awareness Message (CAM) transmission rate in the ITS control channel. Though higher rate is better for control, but it may not be reached when the channel is busy due to a large number of cars on a busy road. The proposed CACC closed loop system model with network delay is described in the next section.

IV. CLOSED LOOP PLATOON SYSTEM

It has been assumed that vehicle at $i = 0$ is a virtual reference vehicle given a reference acceleration a_0 which is followed by rest of vehicles in the string. This is done to ensure vehicle at $i = 1$ to acts as leader and employ same controller as rest of the platoon members. Based on above assumption virtual reference vehicle model can be formulated as:

$$\dot{x}_0(t) = A_0 x_0(t) \quad (18)$$

where $A_0 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ and $x_0 = [q_0 \ v_0 \ a_0]^T$. Equation (14) is for i^{th} vehicle, so for each vehicle in the subsystem following equations can be obtained,

$$\begin{aligned} \dot{x}_1(t) &= A_1 x_1(t) + B_{s,1} u_{1,des,j}(t) \\ &\vdots \\ \dot{x}_n(t) &= A_n x_n(t) + B_{s,n} u_{n,des,j}(t) \end{aligned} \quad (19)$$

By introducing new state vector $\theta_n = [x_0^T x_1^T \dots x_n^T]^T$, and substituting $u_{i,des,j}$ from (15), (16) and (17) in (19). Equations (18) and (19) can be lumped into a single system that can be written in the state space representation for three given topologies as,

$$\dot{\theta}_n(t) = A_{\theta_n} \theta_n(t) + B_{\theta_n,j} u^*(t) + f_{\theta_n}, \quad (20)$$

$j \in \text{PF, PPF, LF}$

where u^* is the control input (* denotes the data obtained via network), A_{θ_n} is system matrix and $B_{\theta_n,j}$ is input matrix that varies with each topology, f_{θ_n} is the feed-forward gain given by,

$$A_{\theta_n} = \begin{bmatrix} A_0 & 0 & 0 & \dots & 0 \\ 0 & A_1 & 0 & \dots & 0 \\ 0 & 0 & A_2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & A_n \end{bmatrix}$$

$$B_{\theta_n,PF} = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 & 0 \\ B_{s,1}k_{1,0} & B_{s,1}k_{1,1} & 0 & \dots & 0 & 0 \\ 0 & B_{s,2}k_{2,1} & B_{s,2}k_{2,2} & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & B_{s,n}k_{n,n-1} & B_{s,n}k_{n,n} \end{bmatrix}$$

$$B_{\theta_n,PPF} = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 & 0 \\ B_{s,1}k_{1,0} & B_{s,1}k_{1,1} & 0 & \dots & 0 & 0 \\ B_{s,2}k_{2,0} & 0 & B_{s,2}k_{2,2} & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & B_{s,n}k_{n,n-2} & 0 & B_{s,n}k_{n,n} \end{bmatrix}$$

$$B_{\theta_n,LF} = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 & 0 \\ B_{s,1}k_{1,0} & B_{s,1}k_{1,1} & 0 & \dots & 0 & 0 \\ B_{s,2}k_{2,0} & 0 & B_{s,2}k_{2,2} & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ B_{s,n}k_{n,0} & 0 & 0 & \dots & 0 & B_{s,n}k_{n,n} \end{bmatrix},$$

$$u^* = [\theta_n] \text{ and } f_{\theta_n} = \begin{bmatrix} 0 \\ B_{s,2}k_{c,2} \\ B_{s,3}k_{c,3} \\ \vdots \\ B_{s,n}k_{c,n} \end{bmatrix}$$

The above system being implemented with a zero-order hold mechanism gives [17],

$$\theta_{n,k+1} = \phi \theta_{n,k} + \Gamma_j u_k^* + f_{\theta_n,k}, \quad (21)$$

$j \in \text{PF, PPF, LF}$

where $\phi = e^{A_{\theta_n}h}$, $\Gamma_j = \int_0^h e^{A_{\theta_n}s} B_{\theta_n,j} ds$, h is the sampling period and $u_k^* = [\theta_{n,k}]$ is the CACC input sent over wireless network.

A. CACC Network Delay model

As we described earlier, the imperfection introduced by the network induces a *time-delay* in the reception of the packets exchanged over the network. At every sampling instance each vehicle transmit its present state via network to all the followers in the form of packets. Each follower receive these packets within the next 100ms time frame. At the end of each sampling period all vehicles simultaneously update their states using these packets having at most 100ms old state information of predecessor, pre-predecessor or the leader vehicle. Hence equation (19) can be rewritten as,

$$\theta_{n,k+1} = \phi \theta_{n,k} + \Gamma_j u_k^* + f_{\theta_n,k} \quad (22)$$

$j \in \text{PF, PPF, LF}$

where $\phi = e^{A_{\theta_n}h}$, $\Gamma_j = \int_0^h e^{A_{\theta_n}s} B_{\theta_n,j} ds$ and $u_k^* = [\theta_{n,k-1}]$. Here u_k^* is CACC input obtained through wireless network delayed by 100ms. Γ_j is the input matrix that varies with each topology. In the next section we will analyze string stability and later validate the analysis results through simulations.

B. String Stability

In this model equilibrium state of virtual reference vehicle ($i=0$) is marginally stable i.e., we assume that the virtual reference vehicle provides the right reference acceleration a_0 with appropriate lower level controller in place. Stability of rest of the platoon members ($i = 1, 2, 3, 4$) has been analyzed using Lyapunov stability criterion [16]. For a given acceleration input a_0 and parameters shown in Table 1,

equation (22) after substituting u_k^* and introducing a new state vector $z_k = [\theta_{n,k}; \theta_{n,k-1}]$ can be rewritten as,

$$z_{k+1} = \alpha_j z_k + C \quad (23)$$

where $\alpha_j = \begin{bmatrix} \phi & \Gamma_j \\ 1 & 0 \end{bmatrix}$ and $C = \begin{bmatrix} f_{\theta_{n,k}} \\ 0 \end{bmatrix} = \text{constant}$. For each topology we obtained six eigenvalues. The absolute eigenvalues $\lambda_{i,PF}$, $\lambda_{i,PPF}$ and $\lambda_{i,LF}$ ($i \in 1, 2, \dots, 6$) corresponding to α_{PF} , α_{PPF} and α_{LF} were found to be less than unity which implies that all three control strategies dependent on communication topologies are stable. $|\lambda_{max,j}| = 0.9677 < 1$, where $j \in PF, PPF$ and LF . Hence, the overall string running under any topology is asymptotically stable.

C. Simulation Results

Fig. 3 shows simulation results for three different topologies using parameters shown in Table I. Fig. 3a and 3b shows the acceleration and velocity profile respectively of all the platoon vehicles in response to a sinusoidal acceleration input (a_0) to the virtual reference vehicle between -2 m/s^2 to 2 m/s^2 until 47 sec and zero acceleration input thereafter.

Fig. 3c shows the the gap variation between vehicles

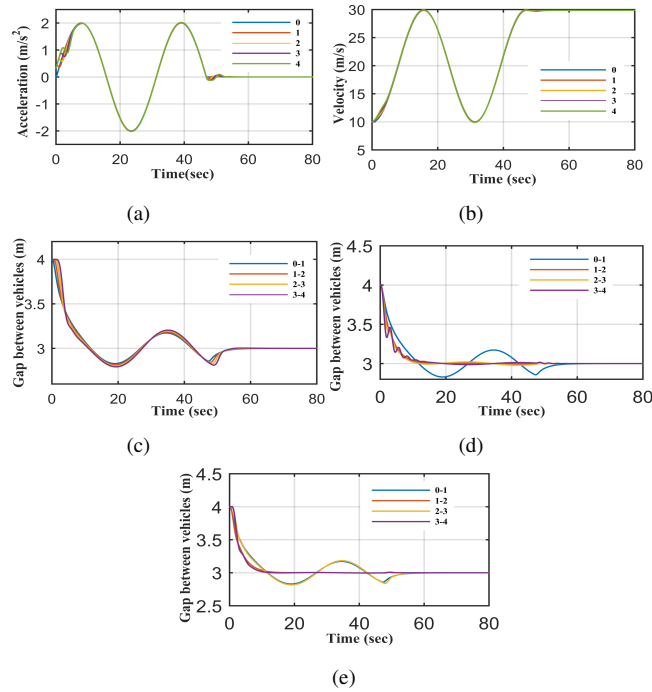


Fig. 3: a) Acceleration Profile b) Velocity profile, c) Gap between vehicles in PF, d) Gap between vehicles in LF, e) Gap between vehicles in PPF.

following PF topology. The reason that each vehicle communicates with its predecessor to perform the vehicle following objective, the gap variations between the vehicles are very much alike. It can be seen that in this topology the disturbance during dynamic conditions propagates throughout the string. The maximum deviation from an ideal gap of 3m can be seen between 3 and 4 which is almost $\pm 6.8\%$.

However Fig. 3d shows the gap variation between vehicles following reference vehicle i.e. LF topology. We can see that the maximum deviation in gap is between 0 and 1 which is almost $\pm 5.7\%$. However the gap between all other vehicles is close to ideal 3m. Each vehicle follows reference vehicle, so any disturbance in reference vehicle is reflected in all the other platoon members at the same time. Thus the relative gap between all other vehicles (1-2, 2-3, 3-4) are close to ideal. Though LF topology appears to be stable and safe. However, as the vehicle gets farther from the leader the probability of receiving its message gets lower. In PPF topology (Fig. 3e) during dynamic conditions the gap between 1-2 and 3-4 are close to ideal similar to LF topology since 2 and 4 are dependent on information from reference vehicle at position 0. The gap between 0-1 and 2-3 varies in the same fashion similar to PF topology. The maximum deviation can be seen between 2-3 which is almost $\pm 6.1\%$.

The above results are obtained based on the assumption of an imperfect communication network with a *time-delay* of 100ms in information exchanged. However this may not be the case always. Under heavy network load condition, probability of packet collision is generally higher. Hence apart from getting delayed, packets can be occasionally lost. To address this, we have integrated all the above topologies in a control algorithm such that each vehicle independently switch between multiple controller strategies based on packets received from the vehicles in-front i.e., either from predecessor or pre-predecessor or the leader. The proposed strategy is described in the next section.

V. PROPOSED CONTROL STRATEGY

In case of three control strategies (described in previous sections) utilizing PF, LF and PPF communication topologies to exchange information, each vehicle is dependent on the information from one of the vehicle in-front. This pattern of information exchange can severely impact the string stability in a congested network when the probability of packet loss is high. Richer information communication between vehicles in CACC opens up the possibility of improving the performance of control algorithm under such conditions. Each vehicle receives information from multiple vehicle in-front. Based on packets received from either the predecessor or pre-predecessor or the leader, each vehicle in the platoon can switch between these three control strategies to obtain a stable and safe operation. Fig. 4 and algorithm 1 further explains the control strategy.

If a vehicle has at least one message received in its buffer it is referred to as an *ideal sample*. On the contrary, if no message is received it is referred to as a *non-ideal sample*. Loss of message is the result of packet collision or delay in message due to congested network or heavy load conditions. Thus there may be instances when messages from all vehicles in-front are lost that too for consecutive periods. In order to embed such real-time communication setting between vehicles, we have considered a probabilistic network model as shown in Fig. 5.

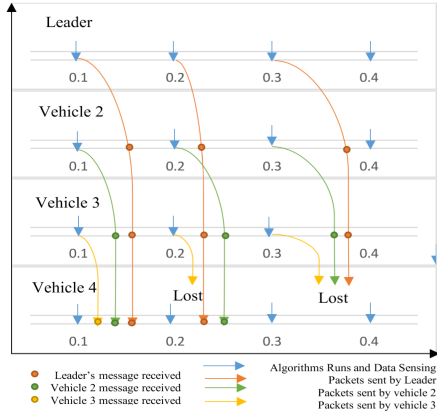


Fig. 4: Control Algorithm and Switching strategy

Algorithm 1 Control Strategy (Between 0.1 sec to 0.2 sec)

- 1: At 0.1 sec, every vehicle computes and update its states based on states at $t=0$
- 2: At 0.1 sec, new states are sensed, and ready to be exchanged in form of packets among platoon members via network
- 3: Between 0.1 sec to 0.2 sec packets are received, each message (packet) is stored in a buffer matrix maintained by each vehicle along with the time stamp of receipt
- 4: At 0.2 sec, each vehicle checks its buffer matrix, for example vehicle 4 in Fig. 4 receives message from leader, vehicle 2 and vehicle 3
- 5: Priority is given to message of predecessor for safety than to Pre-Predecessor and last to leader. In the above example vehicle 4 computes its states with respect to message from vehicle 3, thus follows PF topology
- 6: At 0.2 sec, each vehicle computes and update its states based on message received in buffer matrix
- 7: At 0.2 sec, new states are sensed, and ready to be exchanged in form of packets among platoon members via network
- 8: The process is repeated

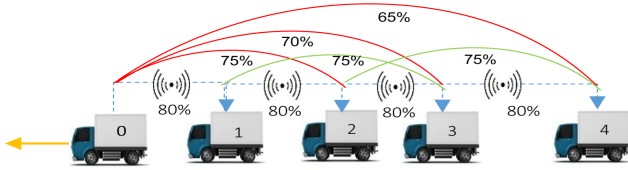


Fig. 5: Communication settings with multiple topologies

In this model we have taken an assumption of an extremely congested network such that probability of receiving a packet from predecessor is 80%, which falls by 5% with subsequent increase in distance from the target vehicle. This model also includes periods of consecutive packet loss (*ideal* and *non-ideal samples* as well). Table II gives the percentage of multiple consecutive misses by each vehicle for 800 samples at a sampling rate of 10 Hz.

Consecutive Period of Missed Packets	1	2	3	4
1	19.80%	4.70%	0.90%	1.50%
2	1.80%	0.10%	0	0
3	0.50%	0	0	0
4	0	0	0	0
5	0.05%	0	0	0
6	0	0	0	0
At least 1 packet received	77.85%	95.20%	99.10%	98.50%

TABLE II: Percentage of consecutive period of packet miss

A. Ideal Samples

In Fig. 4 at 0.3 sec vehicle 4 has two messages in its buffer, one from the leader and other from vehicle 2, while message from vehicle 3 is lost or is delayed by more than 100ms. Based on priority vehicle 4 selects message from vehicle 2 i.e., an *ideal sample*, thus update its states. In this way a vehicle 4 switches from PF to PPF topology. The control scheme for an *ideal sample* make use of both feedback and feed-forward control which is given by equation (22). Thus it can be rewritten for i^{th} vehicle as,

$$x_{i,k+1} = \phi' x_{i,k} + \Gamma_j' u_{i,k,j}^* + f_{x_i} \forall \text{ ideal samples}$$

$j \in \text{PF, PPF, LF}$

where ϕ' is the system matrix, Γ_j' is the input matrix corresponding to the topology selected and f_{x_i} is the feed-forward gain of the i^{th} vehicle. In the above equation $u_{i,k,j}^*$ is the delayed control input obtained via network in the form of packets. Its value depends on message selected by the vehicle based on priority from its buffer i.e., $u_{i,k,PF}^* = [x_{i-1,k-1}]$, $u_{i,k,PPF}^* = [x_{i-2,k-1}]$, and $u_{i,k,LF}^* = [x_{0,k-1}]$. In this case vehicle 4 selects $u_{4,0.3,PPF}^* = [x_{2,0.2}]$.

B. Non-Ideal Samples

In Fig. 4, at 0.4 sec vehicle 4 has no message in its buffer hence a *non-deal sample*. In this situation we have two control schemes. First, by using *feed-forward control only* i.e., nullifying the feedback control input such that,

$$u_{i,k}^* = 0$$

Thus equation (22) can be rewritten as,

$$x_{i,k+1} = \phi' x_{i,k} + f_{x_i} \forall \text{ non-ideal samples.}$$

Second is the *predictive control scheme* in which, a vehicle estimates the position vehicle in-front based on the most recent message received in its buffer history. Selection of which message to choose is done on the basis of priority. The control scheme in this case is given below,

$$x_{i,k+1} = \phi' x_{i,k} + \Gamma_j' \hat{u}_{i,k,j}^* + f_{x_i} \forall \text{ non-ideal samples.}$$

$j \in \text{PF, PPF, LF}$

Here $\hat{u}_{i,k,j}^*$ is the estimated control input ($\hat{\cdot}$ denotes the estimated value). Below is the illustration of the predictive control scheme through the control strategy shown in Fig. 4,

- At 0.4 sec vehicle 4 iterates back and checks its buffer for messages at 0.3 sec.
- At 0.3 sec, vehicle 4 had two messages in its buffer, one from the leader and other from vehicle 2. Based on priority vehicle 4 selects message of vehicle 2 (PPF topology).

- Message contains vehicle 2 states at 0.2 sec i.e., $[x_{2,0.2}] = [q_{2,0.2}; v_{2,0.2}; a_{2,0.2}]$. Vehicle 4 makes an assumption of velocity and acceleration of vehicle 2 to be same as its own velocity and acceleration at 0.3 sec.
- Algorithm computes the displacement of vehicle 2 using: $S = ut + 1/2at^2$, where $u = v_{4,0.3}$, $a = a_{4,0.3}$ and $t = 100ms$.
- Thus estimated position of vehicle 2 at 0.3 sec is given by, $\hat{q}_{2,0.3} = q_{2,0.2} + S$
- Estimated vehicle 2 state at 0.3 sec is given by, $\hat{x}_{2,0.3} = [\hat{q}_{2,0.3}; v_{4,0.3}; a_{4,0.3}]$
- Hence the estimated control input, $\hat{u}_{i,k,PPF}^* = [\hat{x}_{2,0.3}]$
- In this way vehicle 4 computes its new states at 0.4 sec, and the process in Table III is repeated.

At every instance a vehicle selects a communication topology based on packets received in its buffer. In the next section stability of the proposed control strategy utilizing *predictive control scheme* for *non-ideal samples* has been analyzed for an arbitrary switching sequence.

C. String Stability

In the proposed control strategy we have three distinct discrete-time stable LTI systems for three different topologies. Each vehicle perform switching between these systems on the basis of packets received from the vehicle in-front.

A discrete switched linear systems with m stable subsystems described by [18],

$$x(k+1) = A_i x(k) \quad i \in 1, 2, \dots, m \quad (24)$$

where system matrices $A_1, A_2, \dots, A_m \in P^{n \times n}$ are Schur stable (all eigenvalues of a matrix inside the unit circle in the complex plane) $P^{n \times n}$ denotes a set of $n \times n$ dimensional real matrices.

If $A_1, A_2, \dots, A_m$ satisfy simultaneously the following linear matrix inequalities,

$$P - A_i^T P A_i > 0 \quad i \in 1, 2, \dots, m$$

where $P = P^T > 0$ (P is a symmetric positive-definite matrix), then $V(x) = x^T P x$ is a CQLF of discrete switched linear systems (24). Systems (24) are asymptotically stable under arbitrary switching signal.

From (23) Lyapunov equation can be written as,

$$P - \alpha_j^T P \alpha_j > 0, \quad j \in \text{PF, PPF, LF}$$

Solving the above inequality by substituting values of α_j we obtained a common P such that $P = P^T > 0$. Thus a Common Quadratic Lyapunov Function (CQLF) exists for the three systems α_{PF} , α_{PPF} and α_{LF} . Hence proposed switched control strategy is asymptotically stable. Simulation results of the proposed strategy are presented in the next section that further validates the effectiveness.

VI. SIMULATION RESULTS

In this section we will simulate the control algorithm using parameters shown in Table I and initial conditions shown in Table III. Algorithm encompass the communication settings shown in Fig. 5 and Table II along with proposed control schemes for *ideal* and *non-ideal samples*. A sinusoidal acceleration input is given to the virtual reference vehicle between

$-2m/s^2$ to $2m/s^2$ during dynamic conditions until 47 sec, such that its speed oscillates between $10m/s$ to $30m/s$. Acceleration input is brought to zero after 47 sec where velocity of reference vehicle remains constant at $30m/s$. An emergency braking test was also performed to test the effectiveness of the control strategy by bringing virtual reference vehicle to rest from $30m/s$ within 5 sec.

Vehicle	Position (m)	Velocity (m/s)	Acceleration (m/s ²)
0	16	10	0
1	12	10	0
2	8	10	0
3	4	10	0
4	0	10	0

TABLE III: Initial conditions

A. Control Algorithm: feed-forward Control only (for non-ideal samples)

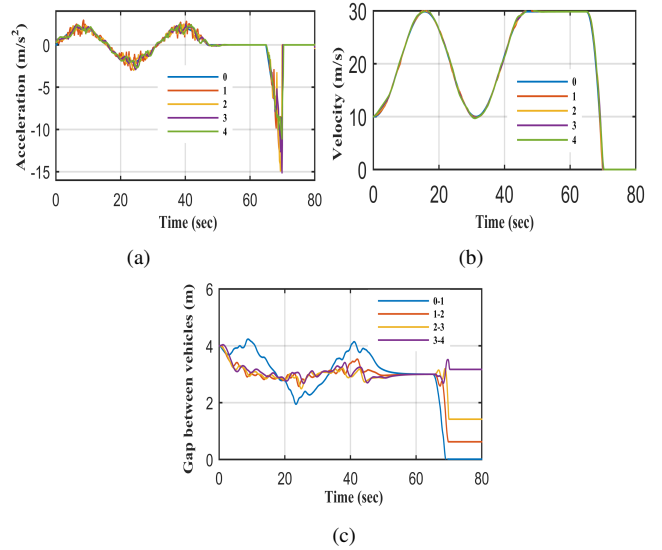


Fig. 6: a) Acceleration profile b) Velocity profile, c) Gap between vehicles

B. Control Algorithm: Predictive control scheme (for non-ideal samples)

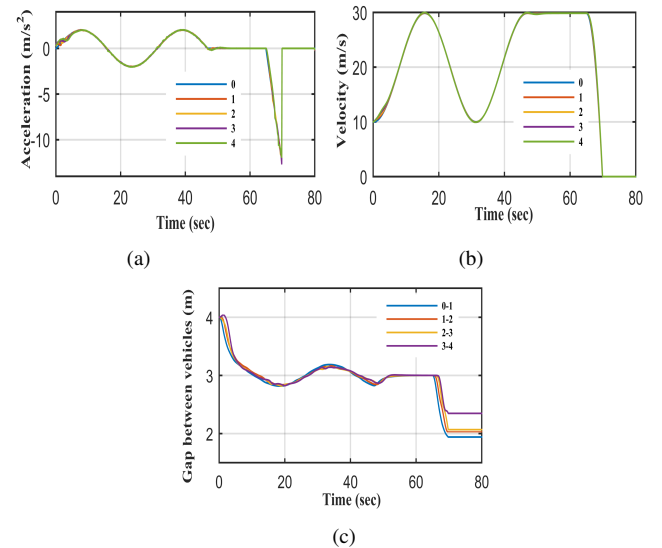


Fig. 7: a) Acceleration profile b) Velocity profile, c) Gap between vehicles

Fig. 6 and 7, shows the simulation results with the switching control strategy and two control schemes during non-ideal samples. In Fig. 6a and 7a, between 47 to 50 sec acceleration of platoon members descends below zero (braking) which is highly unwanted. This is due to the fact in this paper we have an ideal lower level controller. Another reason is the acceleration input given to virtual reference vehicle (0), which is sharply brought to zero.

In Fig. 6a, during dynamic conditions a lot of jitters can be seen. These jitters are the result of switching between topologies and also due to the occurrence of non ideal samples, in which only feed-forward control input is applied. In this situation a vehicle moves forward without any feedback control input until it receives a message from the vehicles in-front. Fig. 6c depicts gap variations between platoon members. Clearly, a maximum deviation from constant buffer gap of 3 m can be seen between 0 and 1, which is almost $\pm 34.9\%$. The deviation is maximum between reference (0) and 1 since each vehicle switches between PF, PPF and LF topologies. Due to LF topology, disturbances are not propagated throughout the chain as shown in Fig. 3c, which compensates for the disturbances that propagates due to PF and PPF topologies. During emergency braking test, collision occurs between vehicle 0 and 1 which confirms the poor performance of this control scheme.

Fig. 7a shows acceleration profile of platoon members using predictive control strategy during non-ideal samples. Here not only jitters are minimized but also this strategy reduces the deviation of gap between members during dynamic conditions. In Fig. 7c maximum deviation from constant buffer gap of 3 m is $\pm 6.3\%$ which is between 0 and 1. For the rest of the platoon members deviance in the gap is less than $\pm 6.3\%$. Emergency braking test performed further confirms effectiveness of this control scheme. The gap between vehicles after braking operation are within the safer limits. Thus compared to control strategy with feed-forward control input only, predictive control strategy shows superior performance by enhancing the stability of the system under imperfect network conditions.

VII. CONCLUSION AND FUTURE WORK

A novel control algorithm has been proposed and evaluated to maintain a stable and safe operation under wide range of network conditions. We have designed multiple control strategies dependent on communication topologies i.e., Predecessor Follower, Pre-Predecessor Follower and Leader follower such that each vehicle switches between these strategies based on packets received from vehicles in-front. Finally the algorithm is evaluated with switching strategy through simulation in Matlab considering message delivery probabilistic model, which shows significant enhancement in the performance and safety improvement for future autonomous driving systems. In this work we have assumed Generalized Vehicular Longitudinal Dynamics for modelling lower level controllers, however for more accurate and practical results proposed control strategy can be evaluated by designing lower level controller with non-linearities

involved due to mass, inertia, aerodynamic drag, friction etc. Further, consideration of fuel consumption, merging and diverging platoons can be an interesting direction of future research.

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